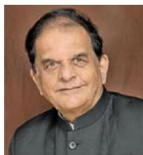


PARALLEL ARRAY OF ESPs HELPS TO HARVEST MORE ENERGY



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P.S. Deodhar set up MITRAMAX Energy after he stepped down from his responsibilities with APLAB. (Mitra is a Sanskrit name for Sun.) He has devoted himself to research in solar energy harvesting technologies and embedded solar panel (ESP) is his company's first product that converts solar panels, whether mono-crystalline or poly-crystalline, into a maximum peak energy tracking high-voltage constant DC power source to make them identical in their electrical characteristics. Any number of ESPs can be hooked in parallel.

ESPs were primarily created for rural maintenance-free solar systems to enable each poor family to get at least 1 kWh energy each day for use at night.

Deodhar speaks to Uma Bansal, deputy editor at EFY, about photovoltaic solar panels and advantages of ESPs

How important is solar power for a developing country like India?

There are indeed several reasons. Primarily, we have a severe power shortage and blackout in most parts of rural India that accounts for two-third of our country. This primary input of societal development can't wait for the grid to reach those unfortunate people in the hinterlands.

Equally, the lack of power leads to poor productivity of rural artisan and farmers. They lose five hours each day when they can create wealth productively, purely because of lack of power after sunset.

Solar power is everywhere; I call it the 'solar grid.' It is nature's bounty and available to every rural farmer and artisan. Fossil fuel based power generation also emits more CO₂ per watt delivered today.

Is India well equipped to reap the full benefits of solar power?

Luckily, harnessing power from solar heat and light energy is a proven and very simple technology. Solar heaters produced in millions would be very useful in harnessing solar heat for water and liquid heating. This alone will save us millions of kilowatt-hours of electrical power.

The same is true about photovoltaic solar panels. The demand is huge and businesses are ready to provide these products. With competition and volume, prices will drop further. Even at current prices, PV solar power costs almost two-third of the cost of power from diesel gensets.

Gensets, in addition to power, generate an alarming amount of noise and air pollution. Replacing them with solar power will lead to a sharp drop in carbon emission and noise that makes people physically sick.

What are the technologies in use for PV cells?

Currently, a photovoltaic cell is the least expensive way to harvest solar electric power. It works at 16-17.5 per cent efficiency and a 161cm² (5-inch²) PV cell delivers 3-4W power at about 0.5V.

Mono-crystalline and poly-crystalline are the two types of cells currently used. Several of these cells are connected in series and sandwiched under specially toughened glass to produce PV solar panels.

Which of the two is more efficient and more economical?

Mono-crystalline offers higher efficiency but since the input energy is free, the cost per watt of peak power determines the choice.

The main problem till now was a loss of power harnessed in the process of connecting the cells in series to obtain a usable voltage level. Unless these cells are matched, each cell in a series string delivers energy as little as the least efficient amongst them. This happens in factories making PV panels from PV cells as well as when PV panels are connected in series, as is always done in PV solar arrays currently deployed.

The loss of power could be as high as 12 per cent unless unit cells and panels are carefully matched when these are connected in series. That's why our technique to process solar energy with an embedded controller inside a PV panel is important for the country.

What is the role of a solar charge controller?

Depending on the strength of solar radiation during a day, a 12V PV panel outputs voltage that varies from about